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RESEARCH ARTICLE

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Structural basis of secreted acid phosphatase polymerization in the *Leishmania* parasite

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Abstract

Enzymes that assemble into filaments typically transition between protomeric and polymeric states in response to cellular conditions. In contrast, the secreted acid phosphatase (SAP) of *Leishmania*, one of the most abundant extracellular glycoproteins produced by the parasite and regarded as a major virulence factor in the neglected tropical disease leishmaniasis, exhibits fundamentally different behavior. Depending on the species, SAP forms either highly stable extracellular filaments or remains exclusively as globular particles, with no evidence of reversible interconversion. This binary assembly pattern is particularly intriguing given that SAP orthologs that differ in their ability to polymerize share a high degree of sequence conservation, leaving the molecular determinants of filament formation unknown. Here, we report the cryo-EM structure of filamentous *Leishmania mexicana* SAP to a global resolution of 3.0 Å. The structure resolves the multilevel organization of the enzyme, from individual catalytic phosphatase domains and their unique substrate-binding pockets to the formation of homodimeric protomers, the decoration with N-linked glycans, and the supramolecular organization into filaments. At the core of the polymerization interface, we identified a unique β-hairpin motif that has not been observed in any other phosphatase or enzyme filament, which provides exceptional filament stability. By integrating structural data with comparative sequence analysis and machine-learning-based structure predictions, we define the molecular basis for the species-specific assembly behaviors observed across *Leishmania* SAPs. This work establishes the principles governing SAP filament formation and provides a framework for understanding its evolution, enzymatic function, and potential applications.

KEYWORDS
acid phosphatase, cryo-EM, enzyme filaments, *Leishmania*

1 | INTRODUCTION

Enzyme filamentation is a widespread and conserved mechanism that regulates catalytic activity and spatially organizes metabolism in cells (Hvorecny & Kollman, 2023; Lynch et al., 2020; Park & Horton, 2019). Although most characterized examples are of intracellular metabolic enzymes (Hunkeler et al., 2018; Johnson & Kollman, 2020; Lynch et al., 2017, 2024; Noree et al., 2019), filament formation is increasingly recognized as a more general organizational principle for enzymes functioning in diverse biological contexts. Such an example is provided by the protozoan kinetoplastid parasite *Leishmania*, which secretes an acid

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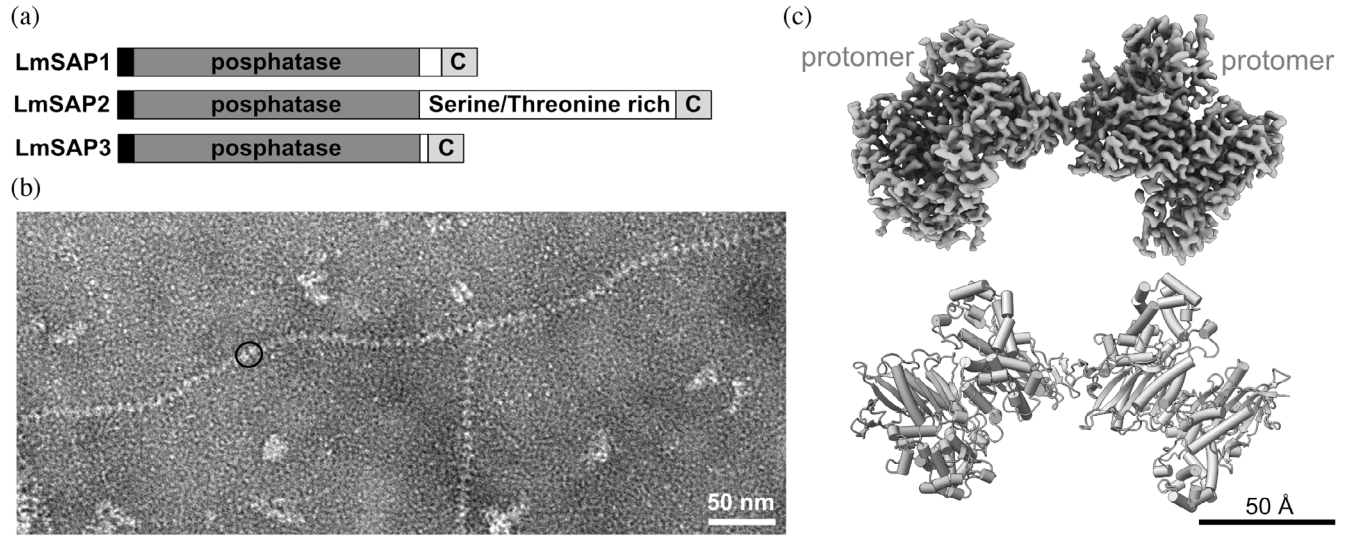


FIGURE 1 Structure of filamentous LmSAP. (a) Domain organization of LmSAP. LmSAP1 and LmSAP2 correspond to published LmSAP sequences (Wiese et al., 1995). LmSAP3 corresponds to the sequence identified in this study (Figure S2b). The signal peptide, the N-terminal phosphatase domain, the Ser/Thr-rich domain, which varies in length among LmSAP proteins, and the C-terminal domain are represented by black, dark gray, white, and light gray bars, respectively. (b) Negative-stain EM of enriched filamentous LmSAP, showing a distinct bead-necklace architecture. The black oval indicates two beads, which correspond to the map and model shown in (c). (c) Cryo-EM map (top) and model (bottom) of LmSAP at pH 7.4, including two beads (protomers). Glycans were removed from the model for simplicity.

phosphatase that assembles into stable, extracellular enzyme filaments in certain species (Ilg et al., 1991).

Leishmania parasites are the cause of leishmaniasis, a globally distributed endemic that the World Health Organization classifies as a neglected tropical disease. Approximately 20 *Leishmania* species infect humans, dogs, and other mammals in nearly a hundred countries worldwide (Baneth & Solano-Gallego, 2022; Costa et al., 2023; Kato, 2025; Niba Rawlings et al., 2025). The parasite alternates between two main life cycle stages: intracellular amastigotes that infect the macrophages of vertebrate hosts, and extracellular promastigotes that inhabit the alimentary tract of the sandfly vector (Bates, 2018). To survive in these radically different environments, the parasites secrete a broad repertoire of phosphoglycosylated macromolecules that mediate host–parasite interactions and adaptation to environmental stress (Elmahallawy & Alkhaldi, 2021; Mule et al., 2020). Among these, secreted acid phosphatase (SAP) is one of the most abundant and conserved extracellular phosphoglycoproteins, present in all *Leishmania* species examined, except for *Leishmania* major (Lovelace & Gottlieb, 1986).

SAP belongs to the histidine phosphatase superfamily (Rigden, 2008). Earlier biochemical studies identified acid phosphatase activity on the cell surface (Fernandes et al., 2013; Freitas-Mesquita et al., 2014; Gomes et al., 2011; Papadaki et al., 2015) and in the culture supernatants of *Leishmania*, with the extracellular enzyme accounting for more than 90% of the total acid phosphatase activity detected in *Leishmania* cultures (Gottlieb & Dwyer, 1982; Hassan & Coombs, 1987; Lovelace & Gottlieb, 1986). SAP is implicated in parasite survival and infectivity through the dephosphorylation of host- and parasite-derived substrates, as well as by modulating host immune responses (Freitas-Mesquita et al., 2021; Gomes et al., 2011; Paulini et al., 2022). Accordingly, SAP is considered a key virulence factor and has been identified as a potential drug target for leishmaniasis (Shang et al., 2024).

Among the filament-forming species, the *Leishmania mexicana* SAP (LmSAP) has been the most extensively characterized. The LmSAP filament comprises two phosphoglycoprotein subunits, namely LmSAP1 and LmSAP2, that assemble into micrometer-long extracellular filaments secreted from the flagellar pocket (Ilg et al., 1991; Stierhof et al., 1994, 1998). LmSAP1 and LmSAP2 both contain a signal peptide, an N-terminal catalytic phosphatase domain, a highly variable Ser/Thr-rich region, and a short C-terminal domain (Wiese et al., 1995) (Figure 1a). Both proteins present N- and O-linked glycans and differ exclusively in the length and sequence of their Ser/Thr-rich regions, which serve as scaffolds for extensive O-linked phosphoglycan modification (Wiese et al., 1995). Either protein is sufficient to generate enzymatically active filaments (Stierhof et al., 1998). Electron microscopy, combined with glycerol spraying and low-angle rotary metal shadowing, revealed that SAP filaments adopt a bead-necklace morphology, with the phosphoglycosylated Ser/Thr arms radiating outward, yielding an overall bottle-brush appearance (Ilg et al., 1991; Stierhof et al., 1994, 1998).

SAPs from different *Leishmania* species exhibit marked differences in composition and assembly. Like

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LmSAP, SAPs from *L. braziliensis* and *L. amazonensis*, representing New World species, form filaments that readily sediment from culture supernatants, whereas SAPs from Old World species, such as *L. donovani*, *L. infantum*, *L. tropica*, and *L. aethiopica*, remain soluble after ultracentrifugation (Ilg et al., 1991). Despite their high sequence similarity, only some SAPs polymerize, raising the question of which molecular determinants facilitate filament formation. Given that the polymerization of other metabolic enzymes is often an evolutionarily accessible trait driven by a small number of surface modifications (Lynch et al., 2020), it is plausible that small sequence or structural differences underlie the distinct assembly behaviors observed among *Leishmania* SAPs; however, their identity remains unknown.

Here, we report the cryo-electron microscopy (EM) structure of *L. mexicana* SAP, providing the first experimental atomic model of this secreted enzyme filament that functions as a virulence factor in the parasite. The structure resolves the hierarchical organization of the filament, from individual subunits and their assembly into homodimeric phosphatase protomers, to glycan decoration, active-site architecture, and higher-order polymerization. The structure reveals a unique polymerization motif not observed in any other known phosphatase or enzyme filament. Together with sequence comparison and machine-learning-based structural prediction analyses, we elucidate the molecular basis of distinct assembly behaviors observed among *Leishmania* SAPs.

2 | RESULTS

2.1 | Production of LmSAP filaments

SAP filaments were initially isolated from wild-type *L. mexicana* cultures; however, the yield was insufficient for structural studies (Figure S1). To obtain adequate protein material, LmSAP was overexpressed in axenic cultures of *L. mexicana* promastigotes. Primers were designed to amplify the entire open reading frame (ORF) of either *LmSAP1* or *LmSAP2* from genomic DNA, as the sequences encoding both genes are identical at the 3′ and 5′ ends (Wiese et al., 1995). Polymerase chain reaction (PCR) amplification yielded a distinct ~1700-bp product, consistent with the expected size of the *LmSAP1* ORF (Figure S2a). The sequence of the PCR-amplified product was similar to that of the *LmSAP1* sequence (Wiese et al., 1995), with some nucleotide mismatches identified, expected to result in a total of 14 amino acid mismatches compared to LmSAP1 and LmSAP2 (Figures 1a and S2b). The discrepancy in sequence may stem from the presence of *L. mexicana* subspecies that were originally isolated from different patients. This sequence, named *LmSAP3*, was obtained from three independent genomic DNA preparations despite using different primers and annealing temperatures, and it was used to overexpress filamentous LmSAP. Episomal overexpression of LmSAP3 in *L. mexicana* promastigotes enabled the recovery of bead-necklace-like filaments from the culture media using ultracentrifugation followed by a sucrose step gradient (Materials and Methods, and Figure 1b). We cannot rule out that the sample included endogenous copies of other LmSAP variants, and therefore, the sample is referred to as filamentous LmSAP hereafter.

To confirm that filamentous LmSAP exhibited catalytic activity, an enzymatic assay was performed over a range of pH values using *para*-Nitrophenylphosphate (*p*NPP) as a substrate. As expected of an acid phosphatase, hydrolase activity was most prominent at pH 4.5–5.2 and decreased with increasing pH values (Figure S3) (Gottlieb & Dwyer, 1982).

2.2 | Structure determination of LmSAP filaments

Cryo-EM and single-particle analyses were performed to determine the structure of filamentous LmSAP at pH 7.4 and 5.6 (Figures S4, S5). LmSAP filaments were clearly observed and traced in cryo-EM micrographs, allowing us to extract particles that contain at least four successive protomers (Figure S4b). Although 2D classification of particles yielded class averages with distinct features (Figures S4c, S5b), ab initio reconstruction failed to generate a 3D volume with the expected shape and resolution. Instead, helical refinement was performed, initially without any helical parameters as input, resulting in a preliminary and satisfactory alignment of the filament segments. The Symmetry Search Utility in cryoSPARC (Punjani et al., 2017a) was subsequently used, identifying a global minimum that indicated a helical rise of 80 Å and a left-handed twist of −164°. Helical reconstruction with these parameters yielded a cryo-EM map that resolved secondary-structure features (Figure S4d). A subsequent round of refinement performed without enforcing symmetry improved the overall resolution of the cryo-EM map, although the map quality decreased toward the edges of the box (Figure S4e). Thus, local refinement was performed on two neighboring protomers within the center of the box, yielding final reconstructions at average resolutions of 3.0 Å at pH 7.4 and 3.7 Å at pH 5.6 (Figures 1c, S4f, g, S5c, d, and Table 1). Models of three LmSAP3 chains were built into these maps, with each copy including the N-terminal acid phosphatase domain (Figure 1c). As expected, density corresponding to the signal peptide was not observed in the cryo-EM map, nor were the flexible Ser/Thr-rich and C-terminal domains (Stierhof et al., 1998). Therefore, the interactions between protomers observed in the cryo-EM map are mediated by

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TABLE 1 Cryo-EM data collection, refinement, and validation statistics for LmSAP.

Data collection and processing	LmSAP at pH 7.4	LmSAP at pH 5.6
Magnification	×130,000	
Voltage (kV)	200	
Electron exposure (e^{-}/Å^2)	30	
Defocus range (μm)	−0.75 to −2.0	−1.0 to −2.5
Pixel size (Å)	0.89	
Symmetry imposed	C1	
Initial particle images (no.)	1,527,263	1,011,263
Final particle images (no.)	524,666	29,779
Estimated resolution Map-to-Map (Å) (FSC threshold = 0.143)	3.0	3.7
Map resolution range (Å)	2.8–3.1	3.2–4.0
Refinement		
Initial model used	AlphaFold prediction of *L. Mexicana* SAP1	LmSAP at pH 7.4
Estimated resolution Map-to-model (Å) (FSC threshold = 0.5)	3.2	4.1
Map sharpening *B* factor	−91	−105
Model composition		
Non-hydrogen atoms	10,259	10,232
Protein residues	1267	1266
Water molecules	19	0
Ligands		
Phosphate	3	3
N-acetyl glucosamine (NAG)	24	24
Mannose (MAN)	2	2
R.m.s. deviations		
Bond lengths (Å)	0.003	0.003
Bond angles (°)	0.637	0.600
Validation		
MolProbity score	1.52	1.58
Clashscore	7.77	10.17
Poor rotamers (%)	0.19	0.37
Ramachandran plot		
Favored (%)	97.54	97.78
Allowed (%)	2.46	2.22
Disallowed (%)	0	0

the N-terminal acid phosphatase domain, consistent with earlier observations of bead-necklace-like filaments formed by truncated LmSAP versions lacking the Ser/Thr-rich and C-terminal domains (Stierhof et al., 1998).

The LmSAP structures determined at pH 7.4 and pH 5.6 are highly similar, with a root-mean-square deviation of 0.6 Å over 1264 aligned residues (Figure S5e). Owing to its higher estimated resolution, the structure at pH 7.4 is described hereafter, with differences observed at pH 5.6 indicated where relevant.

2.3 | The protomers of filamentous LmSAP are homodimers

The LmSAP filament protomers are acid-phosphatase homodimers (Figure 2a). The twofold symmetry exhibited by LmSAP protomers leads to the assembly of nondirectional filaments, as occurs in other metabolic enzymes that form helical filaments, such as acetyl-CoA carboxylase (Hunkeler et al., 2018; Hvorecny & Kollman, 2023). Acid phosphatases are often complexed in dimers (Anand & Srivastava, 2012), and indeed, a Foldseek (van Kempen et al., 2024) search

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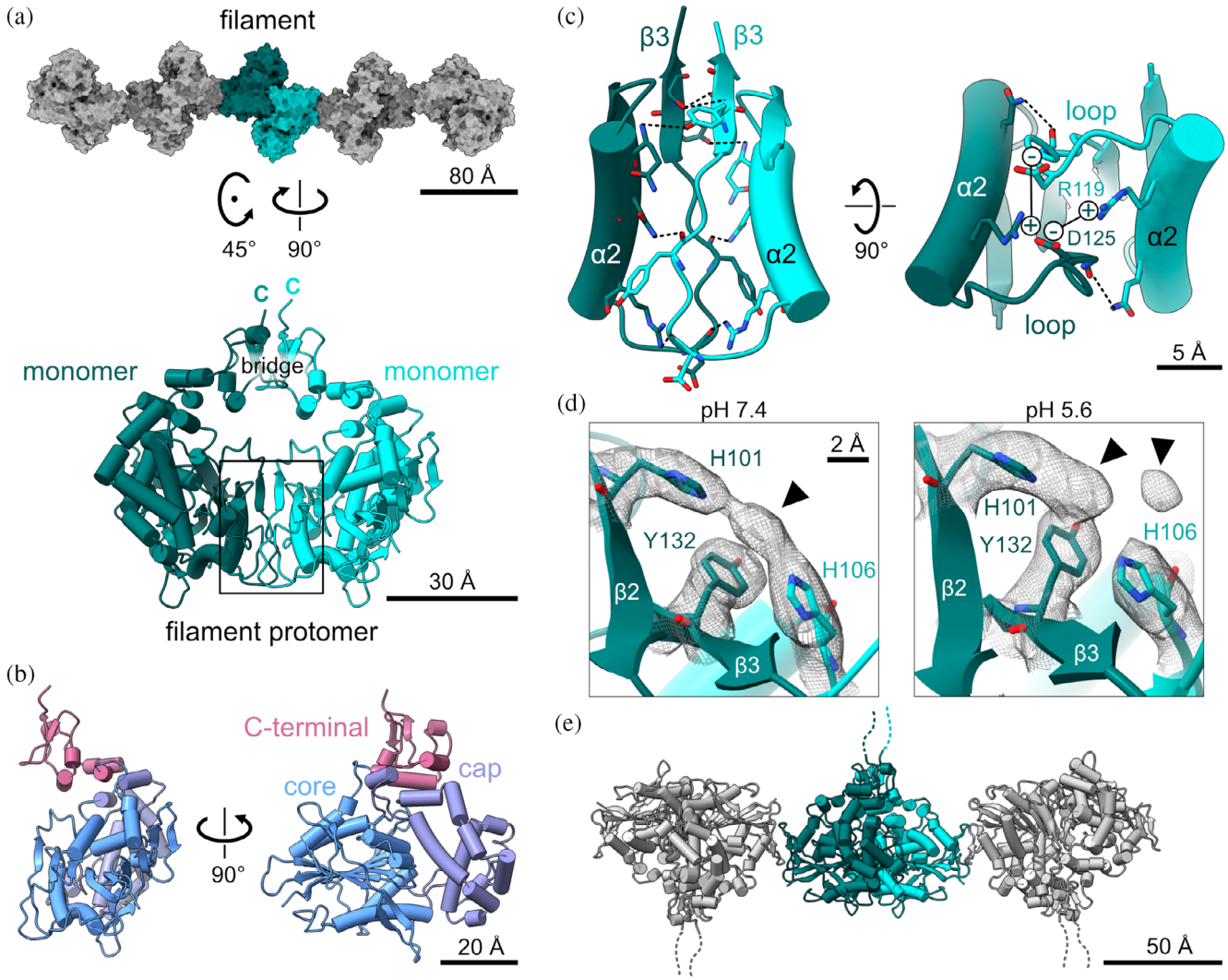


FIGURE 2 The protomers of filamentous LmSAP are homodimers. (a) Model of five successive protomers of filamentous LmSAP with the central protomer colored (top). The same protomer is magnified and shown in cartoon presentation with one monomer colored in cyan and the other in teal (bottom). The C-termini positions of each monomer are indicated. The region created by dimerization of the C-terminal modules is labeled as "bridge." The black frame encompasses the dimer interface formed by the core module, as shown in detail in (c) and (d). (b) The cyan monomer shown in (a) is colored according to its three constituent modules. (c) Zoom into the dimer interface formed by the core module. Left—interactions of the β-strand β3 and α-helix α2 (teal) with their counterparts in the other monomer (cyan). Right—the interface is rotated 90° to show interactions of α2 and the nearby loop with their equivalents in the other monomer. Hydrogen bonds are shown as dashed black lines. Salt bridges (between R119 and D125) are indicated with −/+ signs. (d) Zoom into a unique point of interaction between the monomers of dimeric LmSAP. The cryo-EM map in the vicinity of H101, Y132, and H106 of the neighboring monomer is shown in gray mesh. An unknown density links the histidine residues from the two monomers (indicated with an arrowhead) at pH 7.4 (left) and is altered at pH 5.6 (right). (e) Model of three successive protomers shown in cartoon presentation. Dashed lines indicate the positions from which the Ser/Thr-rich domains are expected to emerge.

for protein structures resembling the structure of LmSAP in the Protein Data Bank identified several structures of acid phosphatase homodimers from both bacteria and mammals (Table S1). Like other histidine phosphatases, the LmSAP phosphatase monomer comprises a core module of a twisted, 7-stranded β-sheet flanked by α-helices and an α-helical module that caps the core, known as the cap module (Dhatwalia et al., 2015; Guo et al., 2021; Lindqvist et al., 1994; Ortlund et al., 2003; Rigden, 2008; Singh et al., 2009). Unique to LmSAP, the C-terminal region of the phosphatase domain (aa 389–447) forms a third module that is stabilized by three disulfide bonds (Figure 2b). Each of these three modules contributes to the dimerization interface of two LmSAP phosphatase monomers within a filament protomer.

The dimer interface area within an LmSAP protomer is at least 1453 Å^2 and is formed by several elements, as estimated using the PDBePISA server (Krissinel & Henrick, 2007). In the center of the dimer interface, the β-strands β3 from the core modules of each monomer interact in parallel via hydrogen bonds (Figure 2c). This

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dimerizing element is common to other acid phosphatase homodimer interfaces (Dhatwalia et al., 2015; Guo et al., 2021; Lindqvist et al., 1994; Ortlund et al., 2003; Singh et al., 2009). A second dimerizing element within the core module is formed by the α-helix α2 and the loop that links α2 with β3. The α2 helix in each monomer interacts with the loop in the opposite monomer through hydrophobic interactions, hydrogen bonds, and two salt bridges formed by R119 and D125 from each monomer (Figure 2c). Another potential dimerizing element within the core module is formed by two histidine residues in β-strand β2 and Y132 in β3 (Figure 2d). In the cryo-EM map at pH 7.4, a distinct density bridges histidine residues 101 and 106 from opposing monomers. At this pH, which is above the pKa of histidine side chains (~6.5), these residues are expected to be predominantly deprotonated and therefore capable of coordinating a cation. The aromatic ring of Y132 is positioned parallel to this density, suggesting the presence of a potential π–cation interaction. Although the resolution of the cryo-EM map does not permit unambiguous identification of the species responsible for this density, its well-defined appearance suggests that it contributes to stabilization of the dimer interface at pH 7.4. At pH 5.6, below the histidine pKa, H101 and H106 are expected to be protonated and positively charged. Consistent with this prediction, the corresponding cryo-EM map shows an altered density distribution, with the unassigned density shifted toward H101, accompanied by an additional, spatially separated globular density near H106. These densities may correspond to water molecules or bound ions.

The cap module contributes to the dimerization interface through the loop linking β-strand β1 and α-helix α1, which forms Van der Waals interactions and hydrogen bonds with the polypeptide chain and an N-linked glycan in the loop connecting α1 and β2 in the neighboring monomer (see below). Finally, a unique dimerization element in LmSAP is formed by the C-terminal modules of each monomer, which together create a "bridge" over the core modules, bringing the C-termini of the phosphatase domains into proximity. The bridges of successive protomers are oriented in opposite directions (Figure 2e), perhaps guiding the phosphoglycosylated Ser/Thr-rich domains emerging from the bridges of each protomer to alternate sides of the filament, consistent with the bottle-brush appearance seen in earlier EM micrographs after glycerol spraying/rotary metal shadowing (Stierhof et al., 1998). This alternating arrangement may prevent steric clashes between the chains emerging from adjacent filament protomers.

2.4 | The LmSAP acid phosphatase domain harbors five N-glycosylation sites

The cryo-EM map presented extra density forming dead-end protrusions near five asparagine residues in each LmSAP3 monomer, specifically N46, 81, 96, 135, 245 (Figure 3). The density could not account for the polypeptide backbone or side chains, indicating the presence of N-linked glycans. Consistent with the cryo-EM map density, these five asparagine residues are within canonical N-glycosylation sequons—NXS/T, where X is any amino acid except P (Figures 3c, and S2b). Within LmSAP3, N219, N410, N450, and N474 are part of such sequons as well (Figure S2b); however, clear density was not observed near asparagine residues 219 and 410, and N450 and N474 are not visible in our model. Mass spectrometry (MS) confirmed the presence of N-linked glycans on N46, 81, 96, 135, 219, and 245 (Materials and Methods and Table S2).

Earlier mass spectrometry and nuclear magnetic resonance studies revealed that LmSAP filaments present two types of N-linked glycans, namely $Glc_1Man_6GlcNac_2$ and $Man_6GlcNac_2$, where Glc is glucose, Man is mannose, and GlcNac is N-acetylglucosamine (Ilg et al., 1994). Our mass spectrometry analysis confirmed that the glycans linked to N46, 81, 96, 135, 219, and 245 comprise two N-acetylhexoseamine sugars and 4–7 hexose sugars (Table S2), consistent with earlier results. The cryo-EM map densities extending from the asparagine residues could accommodate up to three sugars, which correspond to two N-acetylhexoseamine sugars and one hexose. Thus, the trisaccharide $ManGlcNac_2$, or just one or two GlcNac sugars, were modeled and linked to each of the five asparagine residues (excluding N219), depending on the extent of the map density observed in their vicinity (Figure 3b). Glycan trees beyond the first three sugars were not visible in the averaged cryo-EM map, presumably due to their flexibility.

The five N-linked glycosylation sites are distributed over the surface of the filament protomer (Figure 3a). The glycans linked to N81 and 245 are positioned within the core and cap modules, respectively, and do not interact with the protein surface but rather protrude outwards. In contrast, the GlcNac sugars linked to N46, N96, and N135 interact with the nearby protein surface (Figure 3b). The first and second GlcNac sugars linked to N46, which are positioned in the cap module, interact with a neighboring loop within the same polypeptide chain (aa 186–190) via hydrogen bonds, thereby stabilizing the cap module (Figure 3a,b). The glycans linked to N96 and N135 interact with the protein surface of the other phosphatase monomer, thereby contributing to the homodimer interface. Specifically, the glycans linked to asparagine residues 96 in the core module of each monomer are located on opposite ends of the homodimer interface and form hydrogen bonds with loops within the cap module of the neighboring polypeptide chain (aa 53–56) (Figure 3a,b). Asparagine residues 135 from both polypeptide chains are found in proximity, such that the linked glycans emerge from below the protomer bridge to opposite directions,

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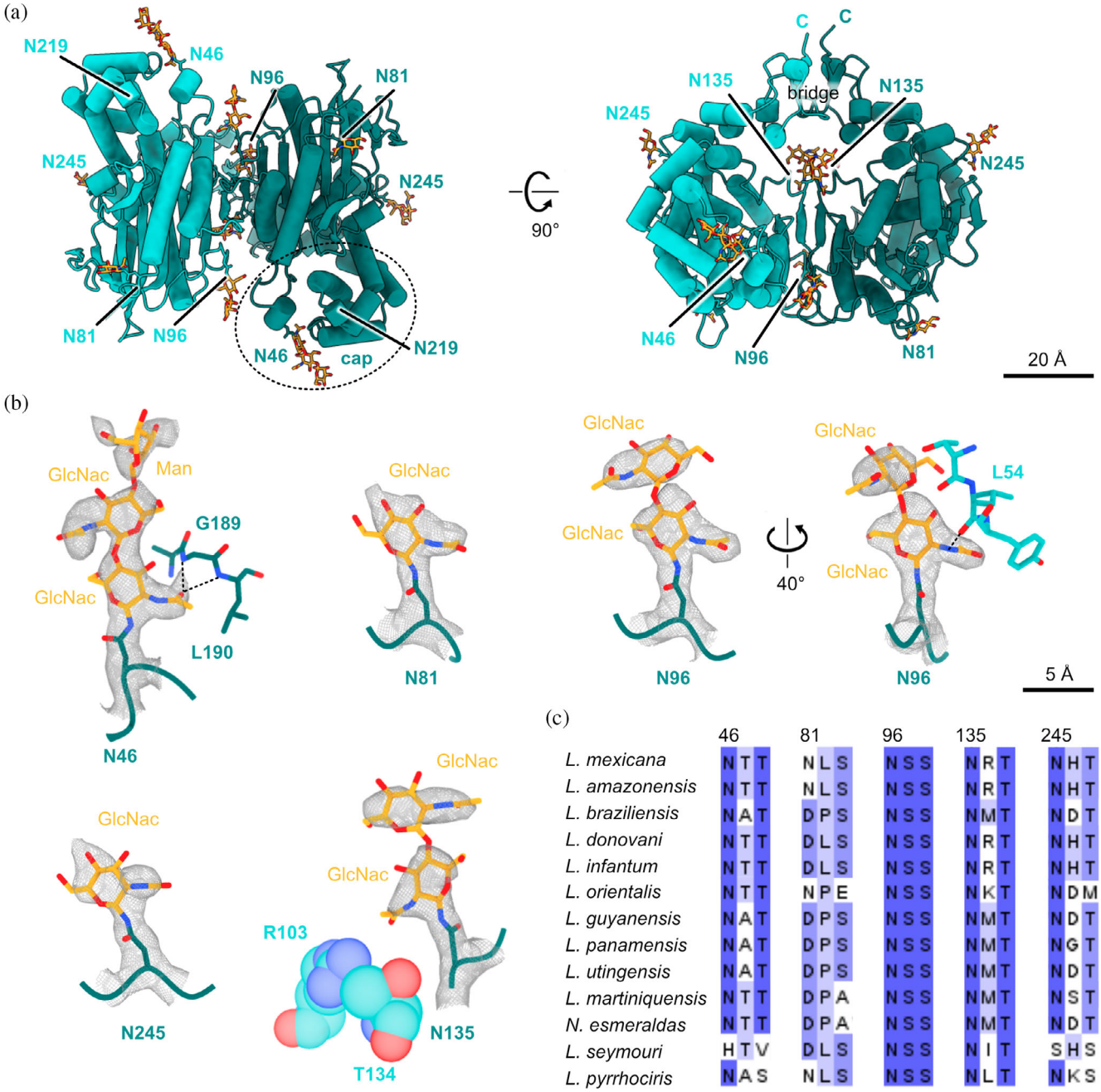


FIGURE 3 The LmSAP acid phosphatase domain harbors five N-glycosylation sites. (a) Two views of a filamentous LmSAP protomer in cartoon presentation with the six N-glycosylation sites (identified by MS) in each monomer indicated. One monomer is colored in cyan and the other in teal. Glycans are presented as orange sticks, except for N219, where clear glycan density was not observed, and the glycan was not modeled. The dashed line circles the cap module in one monomer, where the glycan linked to N46 interacts with the polypeptide chain of a neighboring loop. (b) Cryo-EM maps (gray mesh) and models of glycans (orange sticks) linked to the five asparagine residues (teal sticks). Dashed lines indicate hydrogen bonds. The loop within the cap module that interacts with the glycan linked to N46 is shown in teal sticks. The protein surfaces of the opposite monomer that interact with the glycans linked to N96 and N135 are shown in cyan sticks and spheres, respectively. GlcNac is N-acetylglucosamine and Man is mannose. (c) Multiple-sequence alignment of the five N-glycosylation sequons, for which glycan density was identified in the LmSAP cryo-EM map. Residues are colored according to conservation (dark purple indicates conserved, while white indicates variable). Alignment and visualization were performed using T-Coffee (Notredame et al., 2000) and Jalview (Waterhouse et al., 2009).

thereby interacting with residues 103 and 134 of the other monomer (Figure 3a,b).

The five glycosylated asparagine residues identified here are found in all three *L. mexicana* SAP variants (Figure S2b), and most of them are conserved among *Leishmania* species and related kinetoplastids that encode a SAP (Figure 3c). However, N81 is replaced with aspartic acid in several other species, suggesting that glycosylation at this site is not crucial for *Leishmania* SAP assembly, solubility, or function. The

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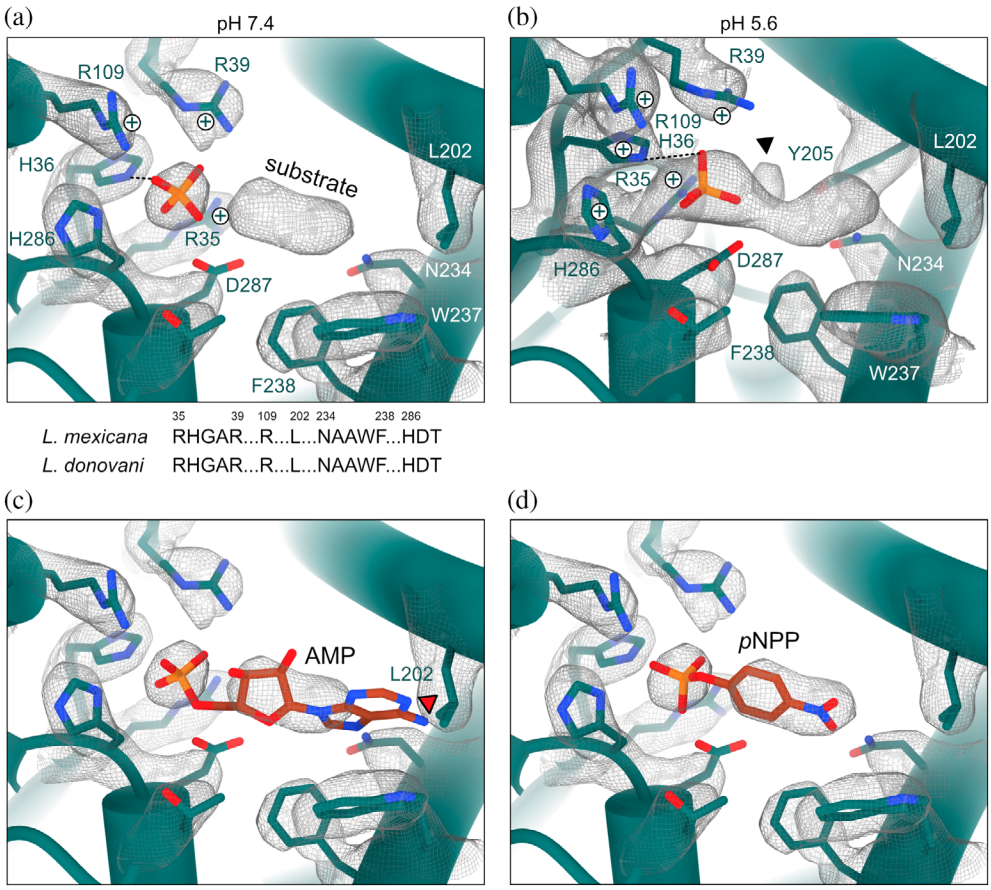


FIGURE 4 Density observed in the LmSAP active site. View of the LmSAP active site at pH 7.4 (a) and pH 5.6 (b). Residues participating in catalysis or substrate binding are shown in sticks. The cryo-EM map in the vicinity of these residues is shown as a gray mesh. Residues interacting electrostatically with the phosphate group in the "phosphate pocket" are indicated with a "+" sign. The phosphate group is shown in orange sticks. A dashed line indicates the transient bond formed between H36 and the phosphate group during catalysis. The density corresponding to the substrate is labeled. A black arrowhead points to an additional globular density observed only at pH 5.6. Sequence alignment of residues forming the active site and substrate-binding pocket shows identity between LmSAP and *L. donovani* SAP. (c) AMP (shown as orange sticks) was modeled into the substrate-binding pocket, however, the adenosine base does not fit the cryo-EM map density and is expected to clash with L202 (indicated with a red arrowhead). (d) *para*-Nitrophenylphosphate (*p*NPP) fits into the substrate-binding pocket, as shown in orange sticks.

sequence containing N96 is the most conserved among the five N-glycosylation sites identified, highlighting its role in stabilizing the dimer interface. Comparison with other high-resolution structures of acid phosphatases, in which N-linked glycans were modeled—namely rat acid phosphatase (Lindqvist et al., 1994) and human prostatic acid phosphatase (Jakob et al., 2000; Ortlund et al., 2003), reveals that the positions of N-glycosylation within acid phosphatase are not conserved (Figure S6), presumably because they do not play a role in enzymatic activity, but rather in protein stabilization or solubility.

2.5 | Density observed in the LmSAP active site

LmSAP belongs to the family of histidine acid phosphatases, and accordingly, possesses an aspartic acid residue at the active site and a "phosphate pocket" (Rigden, 2008). The aspartic acid D287 donates a proton to the leaving group as the substrate transfers its phosphate to the enzyme. The "phosphate pocket" includes arginine residues (R35, R39, R109) and two histidine residues (H36, H286), which interact electrostatically with the phosphate. One of the histidine residues—H36—is located immediately downstream of β-strand β1, and as the catalytic residue, it undergoes transient phosphorylation by the substrate during the catalytic cycle. Density corresponding to a phosphate group adjacent to H36 was observed in the cryo-EM map at both pH 7.4, in which LmSAP is barely enzymatically active, and pH 5.6, in which there is catalytic activity (Figure S3), indicating the presence of a bound substrate, despite the absence of any exogenously added substrate to the LmSAP preparation (Figure 4a,b). At pH 7.4, additional density was observed in the substrate-binding pocket, which was separated from the phosphate density in one of the modeled polypeptide chains. In the two other

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modeled polypeptide chains at pH 7.4 and at pH 5.6, the phosphate and substrate densities were continuous. At pH 5.6, an additional globular density was observed near R35 and Y205 (Figure 4b).

The resolution of the cryo-EM map did not enable the identification of the bound substrate, and therefore, only a phosphate group was modeled in the active site (Figure 4a,b). Given that the enzyme was overexpressed and purified from an axenic culture, this ligand may not represent a physiological substrate. Nevertheless, the structural features surrounding the substrate density offer insight into the substrate-binding properties of LmSAP and the nature of substrates compatible with its active site. The density corresponding to the substrate in the LmSAP structure is relatively small and occupies a distinct position compared to the density observed in earlier structures of histidine phosphatases with the substrate AMP bound (Figure S7) (Guo et al., 2021; Singh et al., 2009). Consistent with this observation, adenosine monophosphate (AMP) does not fit well within the substrate density in the LmSAP cryo-EM map, expected to clash with L202 (Figure 4c), and is a poor substrate of *L. donovani* SAP (Gottlieb & Dwyer, 1982), which is identical to LmSAP in residues forming the substrate-binding pocket (Figure 4a). Smaller phosphorylated molecules that served as better substrates of *L. donovani* and *L. amazonensis* SAPs, such as *p*NPP, glycerolphosphate, and fructose 1,6-diphosphate (Gadelha et al., 2013; Gottlieb & Dwyer, 1982), were more compatible with the substrate-binding pocket (Figure 4d and S8).

The conservation of the residues forming the substrate-binding pocket in LmSAP was next considered. A tryptophan residue (W237) conserved among *Leishmania* species, but absent in histidine acid phosphatases from other organisms, stacks against the substrate density and thus may play an important role in substrate positioning (Figure 4a,b). Additional substrate-surrounding residues, which are largely conserved among *Leishmania* SAPs and likely contribute to substrate binding, are F238, N234, and L202. The enrichment of hydrophobic residues within the substrate-binding site suggests that substrate recognition is largely mediated by hydrophobic interactions.

2.6 | A β-hairpin motif mediates polymerization of LmSAP

In our reconstruction of filamentous LmSAP, adjacent protomers are related by a left-handed rotation of 164° and a translation of 80 Å along the filament axis (Figure 5a). However, as seen in negative-stain EM and cryo-EM micrographs (Figures 1b, S1, S4a), LmSAP filaments may comprise hundreds of protomers and extend as flexible and curved filaments (Ilg et al., 1991; Stierhof et al., 1998), suggesting that not all protomers share a common axis, with some degree of freedom existing within the protomers or at the interfaces between protomers.

In our LmSAP cryo-EM structure, the interface area buried by two adjacent protomers is 560 Å^2. LmSAP polymerization is mediated by three β-strands within the core module, two of which form an anti-parallel β-hairpin, and the third is linked to the β-hairpin via a disulfide bond (Figure 5b). While the disulfide bond is found in globular acid phosphatase structures from diverse organisms (Dhatwalia et al., 2015; Lindqvist et al., 1994; Ortlund et al., 2003; Singh et al., 2009), the β-hairpin motif is exclusive to the structure of filamentous LmSAP, supporting its role in facilitating enzyme polymerization (Figure S9). In contrast to mammalian and bacterial acid phosphatases, sequences of acid phosphatases from *Leishmania* and other related kinetoplastid species include the β-hairpin motif, although not all of them form filaments (Figure 5c). For example, acid phosphatases from *L. donovani* and *L. infantum* are secreted as globular particles, whereas those from *L. amazonensis* and *L. braziliensis* SAP are filamentous, as LmSAP (Ilg et al., 1991). Yet, SAPs from all these species contain a sequence of identical length corresponding to the β-hairpin-forming region, indicating that the presence of this motif alone is insufficient to drive enzyme polymerization.

The inter-protomer interface is formed by a network of hydrogen bonds and hydrophobic interactions (Figure 5d). Specifically, tyrosine 363 is positioned at the center of the inter-protomer interface and stacks against the corresponding residue on the adjacent protomer, forming π–π interactions. Within the same polypeptide chain, the hydroxyl group of Y363 forms a hydrogen bond with N374, while an additional hydrogen bond is formed between Y363 and N357 of the neighboring protomer. Furthermore, the side-chain amide of N361 forms four hydrogen bonds with residues from the neighboring protomer, specifically with the backbone of H366, T367, and I369, as well as the side-chain hydroxyl group of T367, thereby stabilizing the two edges of the protomer interface.

To assess how sequence variation within this region correlates with filamentation, we performed a comparative analysis of the β-hairpin-forming sequences across *Leishmania* species for which assembly behavior has been experimentally characterized. SAPs from *L. donovani* and *L. infantum*, which do not form filaments, share identical sequences in this region and differ from filament-forming SAPs (i.e., *L. mexicana*, *L. amazonensis*, and *L. braziliensis*) at five positions (Figure 5c). Notably, in *L. donovani* and *L. infantum*, Y363 is flanked by arginine residues, whereas in filament-forming species it is typically surrounded by hydrophobic residues such as leucine, valine, or methionine. In addition, N374, which stabilizes the β-hairpin in LmSAP, is replaced by aspartic acid in *L. donovani* and *L. infantum*, potentially introducing electrostatic

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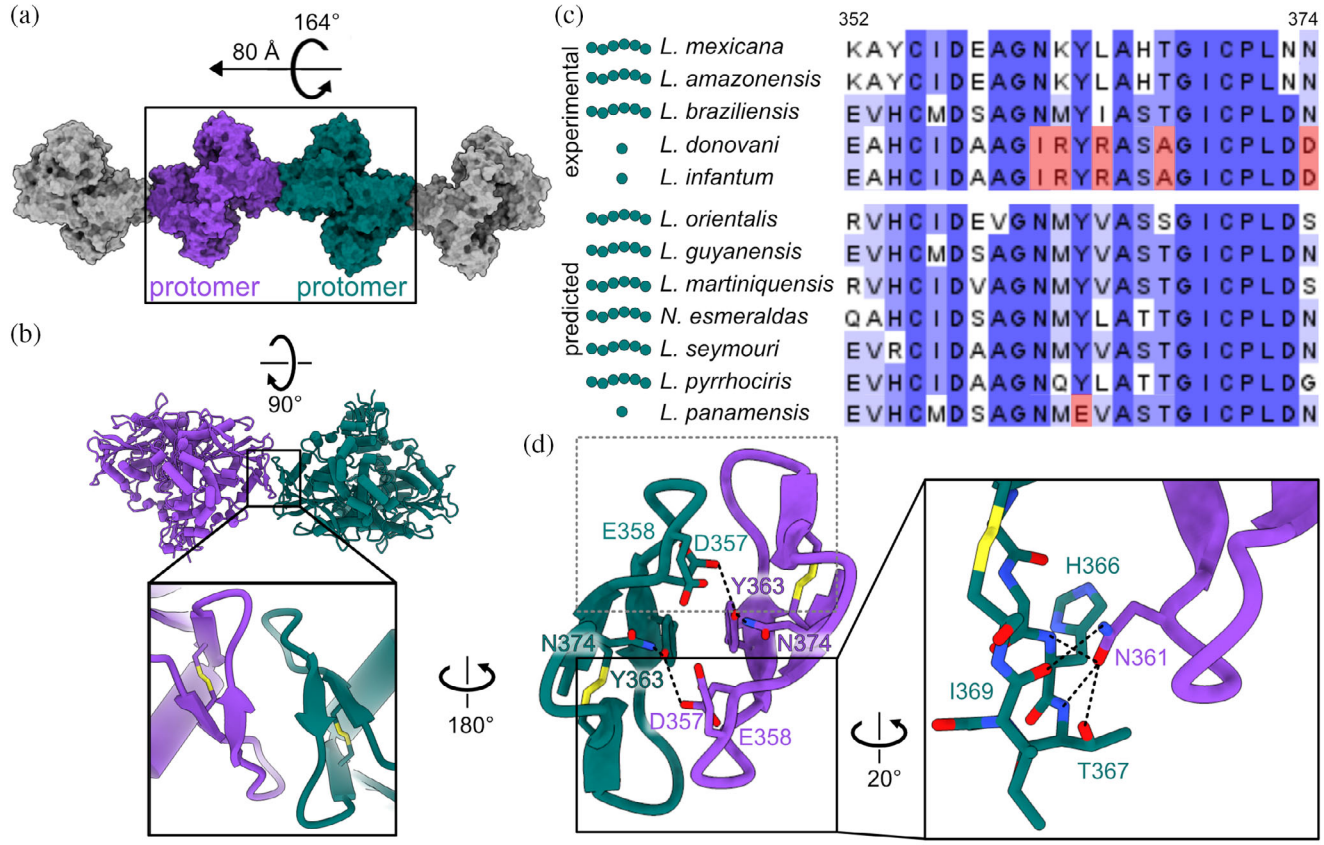


FIGURE 5 A β-hairpin motif mediates polymerization of LmSAP. (a) Model of four successive protomers of filamentous LmSAP with the two central protomers colored in purple and teal. Successive protomers are related by a left-handed rotation of 164° and a translation of 80 Å along the filament axis. The box indicates the two protomers shown in (b). (b) Carton presentation of the protomer interface, which is formed by a β-hairpin motif, stabilized by a disulfide bond (yellow sticks). (c) Multiple-sequence alignment of the β-hairpin-forming region among *Leishmania* and related kinetoplastid species. Top (experimental)—alignment of sequences from species for which assembly behavior has been experimentally characterized. Bottom (predicted)—alignment of sequences from species lacking experimental data. Species that form SAP filaments are indicated with a cartoon of a beaded filament, whereas those that do not form SAP filaments are indicated with one bead (see Figure S10). Residues are colored according to conservation (dark purple indicates conserved, while white indicates variable). Residues within sequences that do not form filaments and differ from the consensus of filament-forming sequences are indicated with a red background. Alignment and visualization were performed using T-Coffee (Notredame et al., 2000) and Jalview (Waterhouse et al., 2009). (d) Close-up of the protomer interface. Left—residues that contribute to the center of the interface are shown in sticks. The boxes indicate the periphery of the interface, stabilized by hydrogen bonds (dashed lines), and the magnified view is shown on the right.

repulsion with E358 across the interface. Similarly, N361 and T367, which form hydrogen bonds in LmSAP, are substituted by isoleucine and alanine, respectively, eliminating side-chain hydrogen-bonding capacity.

To extend this analysis to species lacking experimental data, AlphaFold (Abramson et al., 2024) predictions were generated for SAP sequences from additional kinetoplastids (Figure S10). These predictions correctly classified known filament-forming and non-filament-forming SAPs, indicating that AlphaFold reliably captures features associated with SAP polymerization. Based on these predictions, SAPs from *L. orientalis*, *L. guyanensis*, *L. martiniquensis*, *Novymonas esmeraldas*, *Leptomonas seymouri*, and *Leptomonas pyrrhocoris* are predicted to form filaments, consistent with their sequence similarity to experimentally validated filament-forming SAPs. In contrast, SAP from *L. panamensis* is predicted to remain non-polymeric (Figures 5c and S10). Notably, the β-hairpin region of *L. panamensis* SAP differs from those of filament-forming species at a single position, Y363, following LmSAP numbering. In this species, Y363 is replaced by glutamic acid, which is likely to introduce electrostatic repulsion at the interface rather than stabilizing π–π interactions, thereby disfavoring polymerization.

In conclusion, these observations indicate that SAP polymerization is governed by a combination of sequence-specific interactions within the β-hairpin motif rather than solely by the presence of the motif. To examine whether this polymerization motif is found in other proteins, a structural similarity search was conducted using the β-hairpin-forming regions of two interfacing protomers as input (as shown in Figure 5b,d) in

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the Dali (Holm, 2019) and Foldseek (van Kempen et al., 2024) servers. These searches returned no significant matches, highlighting the uniqueness of this polymerization mechanism implemented by certain *Leishmania* SAPs.

3 | DISCUSSION

Filamentation of metabolic enzymes serves as a regulatory strategy that enables cells to localize, activate, or inhibit enzymatic activity in response to environmental and metabolic cues. In most known systems, polymerization is a dynamic and reversible process, where enzymes transition between protomeric and filamentous states, depending on intracellular conditions such as stress, pH, metabolite levels, or interactions with regulatory proteins (Hvorecny & Kollman, 2023; Lynch et al., 2020; Park & Horton, 2019). The transition between protomeric and filamentous forms often modulates catalytic output, as occurs with acetyl-CoA carboxylase (Hunkeler et al., 2018) and other enzymes, in which filament assembly stabilizes specific active or inactive conformations, allowing rapid switching between regulatory states.

Filamentous *Leishmania* SAP represents an exception to this paradigm. Rather than undergoing reversible transitions, *Leishmania* SAP exhibits an all-or-none assembly mechanism. SAP either consistently forms highly stable filaments in certain *Leishmania* species or exclusively remains as individual protomers in other species (Ilg et al., 1991). To date, there is no evidence that the two states interconvert within a single species. Moreover, SAP filaments are extraordinarily robust, resisting depolymerization under both high and low ionic strengths (1.5 M or 20 mM NaCl), as well as in the presence of detergents such as Triton X-100 and chelating agents, including 10 mM ethyleneDiamine tetraacetic acid (Ilg et al., 1991). This unusual stability suggests that SAP filamentation is not a regulatory switch, but an intrinsic structural property that has been selectively retained in certain *Leishmania* species.

Our cryo-EM structure, along with comparative sequence and structural analyses, reveals the molecular basis of LmSAP filament stability. We identify three main contributors to SAP polymerization: (1) π–π stacking of Y363 with its counterpart from the adjacent protomer, (2) a network of stabilizing interactions formed by residues surrounding Y363 at the center of the protomer interface, and (3) hydrogen bonds formed between loops positioned at the periphery of the interface. Together, these interactions create an extensive interface that accounts for the high stability of LmSAP filaments across diverse chemical conditions. Therefore, the LmSAP polymerization motif is compact yet highly effective, and its modularity suggests that it could, in principle, be harnessed to engineer polymerization into heterologous proteins, as an alternative approach to functionalizing SAP filaments as carriers for recombinant antigens (Kalef, 2021).

Notably, the LmSAP polymerization strategy is not found in any other histidine phosphatase or known enzyme filament. Histidine phosphatases are broadly conserved and typically function as monomeric or dimeric enzymes (Anand & Srivastava, 2012; Rigden, 2008). The emergence of long, extracellular filaments in a subset of *Leishmania* species therefore represents a lineage-specific innovation.

From an evolutionary perspective, SAP filamentation demonstrates how minimal sequence variation can lead to significant changes in protein organization. Despite the high sequence identity among SAPs from different species, our structural and comparative analyses reveal that polymerization correlates with subtle yet functionally consequential differences at the polymerization interface. That such a compact motif can dictate polymerization is consistent with observations across unrelated enzyme systems, where a single surface mutation is often sufficient to induce self-assembly into homomeric polymers (Garcia-Seisdedos et al., 2017, 2019). Thus, divergence in assembly behavior among *Leishmania* species may have arisen from small genetic variations resulting from drift or selective pressures imposed by different sandfly vectors, gut chemistries, or host immune environments.

Why *Leishmania* forms SAP filaments remains an open question. One possibility is that filamentation provides a kinetic or functional advantage. Filament formation may enhance catalytic efficiency by increasing local enzyme concentration, protecting labile substrates or intermediates, or facilitating substrate channeling along the filament (Park & Horton, 2019). Alternatively, SAP filaments may function primarily as structural or immunomodulatory scaffolds, with enzymatic activity playing a secondary role in virulence. A long, highly glycosylated polymer may be better retained and protected in the flagellar pocket or within the digestive tract of the sandfly vector (Stierhof et al., 1998). SAP filaments may also interact with gut surfaces in the sandfly or with the mucin-like proteophosphoglycan matrix produced by the parasite (Ilg et al., 1994; Stierhof et al., 1994; Wiese et al., 1995).

Beyond filament formation, our structure also provides insights into the architecture of the LmSAP protomer. LmSAP retains the canonical features of histidine acid phosphatases, including the core and cap modules, the conserved catalytic histidine, and the positively charged phosphate-binding pocket (Anand & Srivastava, 2012), but exhibits a unique dimerization interface formed by the C-termini, which together create a "bridge" module. This structural addition likely positions the Ser/Thr-rich and C-terminal domains to prevent steric clashes, contributing to the characteristic bottle-brush appearance of the filament (Ilg et al., 1991;

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Stierhof et al., 1994, 1998). The active site is also unusual, accommodating a relatively small and hydrophobic substrate compared to other structurally characterized acid phosphatases (Anand & Srivastava, 2012; Dhatwalia et al., 2015; Guo et al., 2021; Jakob et al., 2000; Lindqvist et al., 1994; Ortlund et al., 2003; Rigden, 2008; Singh et al., 2009). The conservation of active-site residues across *Leishmania* species suggests that enzymatic properties are unrelated to whether SAP polymerizes. Still, the identity of the physiological substrate remains unknown. Finally, our structure mapped the positions of N-glycosylation sites within the acid phosphatase domain, which likely contribute to solubility, extracellular stability, and protection in protease-rich environments encountered during infection (Ilg et al., 1994; Lippert et al., 1999; Wiese et al., 1995).

In conclusion, we define the structural basis for SAP filamentation, revealing a unique polymerization motif that underlies its exceptional stability, and provide a framework for understanding its evolution and potential function. At the same time, our findings are derived primarily from structural analyses of purified filaments and, therefore, the biological significance of SAP filamentation in vivo remains unresolved. Key questions remain, including the identity of the physiological substrate(s) of SAP, the role of enzymatic activity in parasite virulence, and the functions of the Ser/Thr-rich and C-terminal domains. In addition, we provide possible explanations for the species-specific assembly at the structural level, yet the selective pressures that favor filament formation in only a subset of *Leishmania* species remain unclear. Addressing these questions will be essential for uncovering the biological functions of SAP filaments and evaluating their potential as therapeutic targets for leishmaniasis.

4 | MATERIALS AND METHODS

4.1 | Cell growth

Leishmania mexicana strain M379 (MHOM/GT/2001/U1103) cells were cultured in Medium 199 (M199) (Sigma), pH 7.4, supplemented with 10% fetal bovine serum (FBS) (Gibco), 5 μg/mL hemin (Sigma), 0.1 mM adenine (Sigma), 40 mM HEPES, pH 7.4, 4 mM L-glutamine (Sartorius), 100 U/mL penicillin, and 100 μg/mL streptomycin (Sigma) at 26°C.

4.2 | SAP cloning and overexpression

Genomic DNA was isolated from *L. mexicana* promastigote cells with the PureLink Genomic DNA Mini Kit & Tissue Kit (Invitrogen). Fifty nanograms of isolated genomic DNA were used per PCR reaction. Primers for amplification of *LmSAP* from genomic DNA were designed based on the *L. mexicana SAP1* sequence in TriTrypDB (Aslett et al., 2010; Wiese et al., 1995) (accession number: LmxM.36.6480):

5′-CACACGCACCCTCCTCTCTTCCCCTGCCTCCCTCGCACCGGATCCATGGCCTCTAGGCTCGTCCGTGTG-3′ (Forward).

5′-CACGTGGCCGCCCCGCCAGCCGGTGGTCTTCTCGTCCATTCTAGACTGTTGGCGGTGCCGGCTGTACTC-3′ (Reverse).

A PCR product of ∼1700 bp was obtained and cloned into the BamHI/XbaI sites of the pX-based transfection cassette pX-H-SBP-SAP-H (Zinoviev et al., 2011), where H represents the intergenic region of HSP83 genomic locus in *Leishmania* and SBP represents a streptavidin-binding peptide affinity tag of size ∼4 kDa. The cassette-containing plasmid (20 μg) was transfected into *L. mexicana* cells as described previously (Laban & Wirth, 1989). Cells were gradually selected for resistance to G418 (100 μg/mL). Insertion of the cassette and expression of a protein fused to SBP were confirmed using Western Blot analysis.

4.3 | SAP filament purification

LmSAP-overexpressing *L. mexicana* promastigotes were grown in 100 mL of M199 medium in the presence of 4% (v/v) heat-inactivated FBS for 48 h at 26°C to mid-logarithmic growth phase ($2.5–3*10^7$ cells/mL). The culture supernatant was collected (100 mL) following centrifugation for 10 min (4000 g, 4°C) and subjected to ultracentrifugation for 1.5 h (157,000 g, 4°C) using a Sorvall™ WX+ ultracentrifuge. The enzyme-containing pellet was resuspended in 1 mL of lysis buffer (10 mM $MgCl_2$, 100 mM NaCl, and 40 mM HEPES, pH 7.4), layered over an 11-mL sucrose step gradient ranging from 0.5 to 2 M sucrose, and ultracentrifuged overnight at 130,000 g, 4°C. Fractions of 500 μL were collected sequentially from the top of the tube and dialyzed against lysis buffer at 4°C.

For purification of LmSAP at pH 5.6, cells were cultured in the same medium, except that the pH was adjusted from 7.4 to 5.5 using 0.5 M succinic acid. The purification procedure was otherwise identical to that performed at pH 7.4. However, prior to application onto the sucrose gradient, the enzyme-containing pellet was resuspended in 10 mM $MgCl_2$, 100 mM NaCl, and 40 mM sodium acetate pH 5.6, and subsequent dialysis was carried out against the same buffer.

The presence of SAP filaments was evaluated using negative-stain EM and was confirmed in fractions containing 1.45–1.5 M sucrose. For negative-stain EM, 3 μL of sample was applied to glow-discharged 300 mesh copper grids covered by a thin layer of continuous carbon type-B film (Ted Pella) and stained with 2% uranyl acetate. The grids were imaged on a Talos

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F200C microscope (Thermo Fisher Scientific) operated at 200 kV at a nominal magnification of ×22,000 to ×45,000 using a Ceta 16 M pixel CMOS camera (Thermo Fisher Scientific). Micrographs were recorded using Velox (Thermo Fisher Scientific).

4.4 | Phosphatase enzymatic assay

LmSAP phosphatase activity on the substrate *p*NPP was measured in a range of pH values. Purified LmSAP (at a final concentration of 0.1 mg/mL) was mixed with *p*NPP (at a final concentration of 5 mM) in a buffer containing 40 mM Hepes at pH 4.5, 5.2, 6.5, 7.4, or 8.5, 10 mM $MgCl_2$, and 100 mM NaCl at a total volume of 0.2 mL. The reaction mix was incubated at 37°C and quenched after 5 min by adding 0.8 mL of 1 M NaOH. The concentration of the hydrolysis product, *p*NP, was determined by measuring its absorbance at 405 nm, using an extinction coefficient calculated from a calibration curve of pNP at known concentrations and corresponding measured absorbance values at 405 nm. A reaction mix lacking the enzyme was used as a blank for the absorbance measurements. The percentage of consumed substrate was calculated from the initial concentration of *p*NPP placed in the reaction mix and the concentration of hydrolyzed product. Three independent measurements were performed at each pH value.

4.5 | Sample preparation for cryo-EM

Fractions containing LmSAP filaments were concentrated using an Amicon Ultra Centrifugal Filter with a 10 kDa MW cutoff (Sigma) to a volume of ∼30 μL. Three microliters aliquots of purified LmSAP filaments were deposited on glow-discharged Quantifoil R 1.2/1.3 holey carbon grids (Quantifoil, Großlöbichau, Germany). The sample-bearing grids were manually blotted for 4 s at room temperature and vitrified by rapid plunging into liquid ethane using a home-built plunging apparatus. The frozen samples were stored in liquid nitrogen until they were imaged.

4.6 | Cryo-EM data acquisition

Cryo-EM datasets were collected from two independent grids under cryogenic conditions. Grids were loaded onto a Glacios microscope operated at 200 kV and equipped with a Falcon 4i Direct Electron Detector coupled to a Selectris X energy filter (Thermo Fisher Scientific) set at ±5 eV from the zero-loss peak. Movies were recorded in a dose-fractionated counting mode using EPU (Thermo Fisher Scientific) with a pixel size of 0.89 Å and a total electron dose of 30 e−/Å^2. Data were collected at a de-focus range of −0.75 to −2.0 μm (data set at pH 7.4) or −1.0 to −2.5 μm (data set at pH 5.6). Further data collection statistics are reported in Table 1.

4.7 | Cryo-EM data processing

Dose-fractionated image stacks were imported into cryoSPARC (v4.6.2) (Punjani et al., 2017b) and subjected to patch-based motion correction and patch-based contrast transfer function (CTF) estimation. Micrographs with a CTF fit better than 4.5 Å resolution were retained for further processing (5477 micrographs from an initial 14,739 at pH 7.4 and 2381 micrographs from an initial 13,266 at pH 5.6). Particles were picked using the Filament Tracer in cryoSPARC with a separation distance of 80 Å between boxes, corresponding to the estimated size of one protomer within the filament. Particles were extracted at a box size of 342 Å, resulting in an initial number of 1,527,263 (data set at pH 7.4) or 1,011,263 (data set at pH 5.6) particles. The particles were subjected to two-dimensional classification, from which 524,666 (pH 7.4) or 29,779 (pH 5.6) particles were chosen for refinement. To align the particles, helical refinement was initially performed without applying any helical parameters. To determine the spatial relationship between successive protomers along the filament axis, the symmetry search utility in cryoSPARC was used, identifying a global minimum in the mean squared error surface plot that indicated a helical rise of 80 Å and a left-handed twist of −164°. These helical parameters were applied in a second round of helical refinement, yielding a 3D reconstruction at an average resolution of 3.4 Å (pH 7.4). The volume was low-pass filtered to 30 Å and subjected to non-uniform refinement without applying symmetry, yielding 3D reconstructions at global resolutions of 3.0 Å (pH 7.4) and 3.9 Å (pH 5.6). Finally, local refinement was performed on two neighboring beads at the center of the box. A mask was created around the beads of interest, and the signal outside of the mask was subtracted from the particles. The subtracted particles were subjected to local CTF refinement followed by local refinement using a tight mask. The overall resolutions of the resulting maps were estimated at 3.0 Å (pH 7.4) and 3.7 Å (pH 5.6) using the gold-standard Fourier Shell Correlation criterion (FSC = 0.143). For visualization purposes only, the map was sharpened using DeepEMhancer (Sanchez-Garcia et al., 2021). Statistical information for the final density maps is presented in Table 1.

4.8 | Model building

An initial model including three copies of LmSAP1 was generated using AlphaFold (Jumper et al., 2021) and

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was docked into the cryo-EM map obtained at pH 7.4 using UCSF ChimeraX (Goddard et al., 2018). The N-terminal 23 amino acids, which correspond to the signal peptide, and the C-terminal residues 451–537, which correspond to the Ser/Thr-rich and C-terminal domains, were removed from the model. Out of 537 amino acids per chain in this initial model, residues 24–446, 24–447, and 24–443 were retained in chains A, B, and C, respectively. The model was adjusted into the cryo-EM map and mutated to the LmSAP3 sequence using COOT (Emsley & Cowtan, 2004). A model of the N-linked glycan containing two N-acetylglucosamine (GlcNac) units and a mannose (Man) was generated in GlycoGlyph (Mehta & Cummings, 2020). The glycan model (or just one or two GlcNac units, depending on the extension of the cryo-EM map density near each asparagine) was linked to asparagine residues 46, 81, 96, 135, and 245 in COOT using AceDRG (Long et al., 2017). The model was real-space refined against the cryo-EM map using PHENIX (Adams et al., 2010), iteratively rebuilt in COOT, and refined in PHENIX until completion. Model validation and quality assessment, including Ramachandran angles, geometry restraints, clashes, and rotamers, was performed with MolProbity (Chen et al., 2010). The Map-to-Model resolution was estimated at 3.2 Å using the FSC = 0.5 (pH 7.4). This model was docked into the cryo-EM map obtained at pH 5.6 as an initial model, real-space-refined against the cryo-EM map in PHENIX, iteratively rebuilt in COOT, and refined in PHENIX until completion. The map-to-model resolution was estimated at 4.1 Å using the FSC = 0.5 (pH 5.6). Protein–protein interactions were analyzed with PISA (Krissinel & Henrick, 2007). The Foldseek Search Server (van Kempen et al., 2024) and Dali server (Holm, 2019) were used to compare the LmSAP structures to existing structures in the wwPDB. Molecular graphics figures were prepared using UCSF ChimeraX (Goddard et al., 2018).

4.9 | Identification of N-linked glycans using mass spectrometry

Purified LmSAP (60 μL) was mixed with 60 μL of 50 mM Tris–HCl, 10% sodium dodecyl sulfate, and reduced with 5 mM dithiothreitol at 57°C for 1 h, followed by alkylation with 10 mM iodoacetamide at RT in the dark for 45 min. Phosphoric acid was added to the sample to a final concentration of 1.2%, followed by the addition of 90% methanol in 5 mM ammonium bicarbonate. The sample was loaded onto an S-Trap 96-well plate (Protifi, USA) and digested with 0.2 μg of trypsin overnight at 37°C. The digested peptides were eluted using 50 mM ammonium bicarbonate, and trypsin was incubated with the eluent for 4 h at 37°C. Two more eluent fractions were obtained using 0.2% formic acid and 0.2% formic acid in 50% acetonitrile. The three eluent fractions were pooled and vacuum centrifuged. The sample was kept at −20°C.

Ultra liquid chromatography/MS grade solvents were used for all chromatographic steps. The sample was loaded using split-less nano-Ultra Performance Liquid Chromatography (nanoElute2, Bruker Daltonics, Germany). The mobile phase consisted of: (A) 0.1% formic acid in water and (B) 0.1% formic acid in acetonitrile. Desalting of the samples was performed using a reversed-phase PepMap C18 trapping column (300 μm internal diameter, 5 mm length, Thermo Scientific, USA). The peptides were then separated using an Aurora nano-column (75 μm internal diameter, 250 mm length, 1.9 μm particle size; IonOpticks, Australia) at 0.30 μL/min. Peptides were eluted from the column into the mass spectrometer using the following gradient: 2% to 38% B during 60 min, 38% to 95% B during 0.5 min, maintained at 95% for 4.65 min, and then back to initial conditions.

The nanoUPLC was coupled online to a quadrupole time-of-flight mass spectrometer (timsTOF Pro, Bruker). Data was acquired in data-dependent acquisition with ion mobility mode (DDA-PASEF). MS1 range was 100 to 2500 Th. For ion mobility, the 1/K0 range was 0.50–1.60 Vs/cm^2 with a 100 msec ramp time. Fragmentation was performed with stepped CID over 35.0–59 and 40–100 eV. Other parameters were set to the default values of the DDA PASEF method.

Data was searched using Byonic search engine (Protein Metrics, USA) against the LmSAP3 sequence. The glycan database searched against was HexNAc(2) Hex(4–8) based on earlier studies (Ilg et al., 1994). Allowed modifications were fixed carbamidomethylation on C, variable M oxidation, protein N-terminal acetylation, and NQ deamidation. A single glycan was allowed for each peptide. Results were filtered for Pep 2D of |log10| >2. The best identifications for each glycopeptide (glycan-peptide combination) were inspected manually.

4.10 | Prediction of filament-forming potential

The protein sequence of LmSAP1 was used in a BLAST (https://blast.ncbi.nlm.nih.gov/Blast.cgi) search to identify additional SAP sequences from *Leishmania* and related kinetoplastid species. Five copies of each chosen SAP sequence were subjected to structure prediction using AlphaFold (Abramson et al., 2024). The predicted models were evaluated for their potential to form filaments based on the position of interaction between protein chains and the confidence score of these interactions (see Figure S10). Chosen sequences were aligned using T-Coffee (Notredame et al., 2000) and visualized using Jalview (Waterhouse et al., 2009) version 2.11.2.0.

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AUTHOR CONTRIBUTIONS

Priyanka Bose: Conceptualization; methodology; data curation; formal analysis; investigation; writing – review and editing. **Irit Dahan:** methodology; formal analysis; writing – review and editing. **Alexander Upcher:** methodology; writing – review and editing. **Ran Zalk:** methodology; data curation; writing – review and editing. **Iris Grossman-Haham:** Conceptualization; methodology; data curation; validation; supervision; formal analysis; funding acquisition; visualization; writing – original draft.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflicts of interest with the contents of this article.

DATA AVAILABILITY STATEMENT

The cryo-EM maps of LmSAP have been deposited in the Electron Microscopy Data Bank (https://www.emdataresource.org/) under the ID codes: EMD-55773 (pH 7.4) and EMD-56536 (pH 5.6). The atomic models of LmSAP have been deposited in the Protein Data Bank (PDB) (https://www.rcsb.org/) under the accession numbers: 9TBJ (pH 7.4) and 28IK (pH 5.6). The entries listed in Table S1 used in this study were downloaded from the PDB.

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